

Meet Alshi
150096724028

Github : <https://github.com/Meet-6226/Flutter-Assignment/tree/assignment6>

What I learned

While working on this Flutter assignment, I learned how to create a dynamic product listing using `ListView.builder`. Earlier, I mostly worked with static widgets where each item had to be written separately. With `ListView.builder`, I understood that I can store product information in a list and generate the UI automatically for every product. This makes the application easier to manage and also allows it to handle a larger number of products without writing the same code again and again.

I also learned how to create and use a **data model class** in Flutter. I created a `Product` class containing details such as the product name, category, price, rating, image, and description. This helped me understand how real applications organize data instead of keeping every piece of information directly inside the UI code. Using a model also made the product cards cleaner because the UI could simply take the required information from each `Product` object.

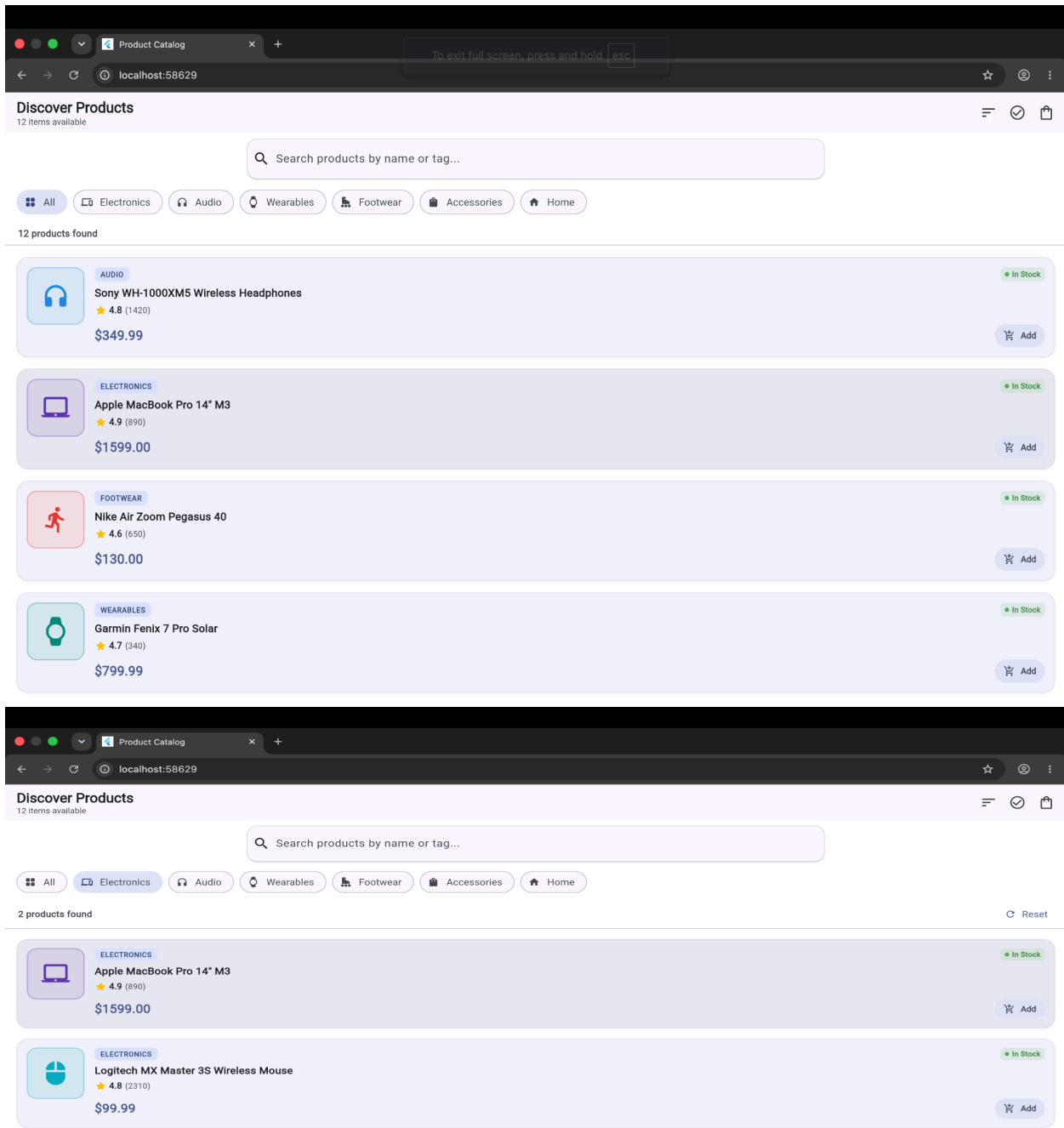
Another important thing I learned was how to use **`StatefulWidget` and `setState()`**. Since the product list changes when the user searches or selects a category, the screen needs to update whenever the data changes. By using `setState()`, I learned how Flutter rebuilds the required part of the interface when a value changes. This gave me a better understanding of how Flutter manages changing data and user interactions.

I also implemented a **search functionality** using a `TextField`. When the user enters a product name or category, the application checks the available products and displays only the matching results. I learned how to listen for changes in the search field and apply filtering logic to a list. I also understood how search and category filters can work together to make the product listing more useful and interactive.

The category filter helped me understand how user selections can affect the displayed data. I added different categories such as Electronics, Fashion, Beauty, Sports, and Accessories, along with an "All" option. When a category is selected, `setState()` updates the selected category and rebuilds the product list. I also added an empty state for situations where no products match the search or selected category, which helped me understand the importance of handling different user inputs properly.

Finally, I learned how to combine Flutter widgets and programming logic to create a more modern and user-friendly interface. I used widgets such as `ListView.builder`, `TextField`, containers, icons, and cards to create the product screen. The main challenge was understanding the filtering logic and making search and category selection work together

correctly. Overall, this assignment improved my understanding of dynamic lists, model classes, state management, and user interaction in Flutter, and showed me how these concepts are used together in a practical application.



Product Catalog

localhost:58629

Discover Products

12 items available

☰

✓

📋

🔍 Search products by name or tag...

☰ All

📁 Electronics

🎧 Audio

🕒 Wearables

👟 Footwear

🛍️ Accessories

🏠 Home

2 products found

🔄 Reset



FOOTWEAR

Nike Air Zoom Pegasus 40

★ 4.6 (650)

\$130.00

🟢 In Stock

🛒 Add



FOOTWEAR

Adidas Ultraboost Light Running Shoes

★ 4.4 (420)

\$190.00

🔴 Out of Stock

🛒 Add

Product Catalog

localhost:58629

Discover Products

12 items available

☰

✓

📋

🔍 Search products by name or tag...

☰ All

📁 Electronics

🎧 Audio

🕒 Wearables

👟 Footwear

🛍️ Accessories

🏠 Home

2 products found

🔄 Reset



WEARABLES

Garmin Fenix 7 Pro Solar

★ 4.7 (340)

\$799.99

🟢 In Stock

🛒 Add



WEARABLES

Samsung Galaxy Watch 6 Classic

★ 4.6 (680)

\$399.99

🟢 In Stock

🛒 Add

